

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fourth Semester of B.Tech. (CE/IT) Examination

May 2011

IT205 Operating System

Date: 12/05/2011

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately

SECTION - I

Q1

- (a) Draw and explain a 5-state process transition diagram.
- (b) What is scheduler? Explain medium-term scheduler.
- (c) Differentiate a physical address and a virtual address.
- (d) What information is saved and restored during a context switch?

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Q2

- (a) Compare paging with segmentation with respect to the amount of memory required by the address translation structures in order to convert virtual addresses to physical addresses?
- (b) Explain working flow of Peterson's solution when interrupt is occurred at the position Other = 1 - Process.
- (c) How do we avoid race condition?

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OR

Q2

- (a) What is semaphore? Which are operations done on semaphore? Give implementation of producer-consumer problem with bounded buffer using semaphore.
- (b) How does device controller interact with CPU by means of interrupt?
- (c) Explain directory structure in UNIX.

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Q3

- (a) Consider a 'claim matrix', an 'allocation matrix' and 'available vector' for a set of processes.

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	Claim Matrix				Allocation Matrix			Available Vector		
	R1	R2	R3		R1	R2	R3	R1	R2	R3
P1	3	2	2	P1	1	0	0	1	1	2
P2	6	1	3	P2	5	1	1			
P3	3	1	4	P3	2	1	1			
P4	4	2	2	P4	0	0	2			

Answer the following questions using the Banker's algorithms:

- i) What are the contents of the matrix need?
 - ii) Is the system in a safe state? If yes, show safe sequence
 - iii) A resource request for one of the processes is given. For example, if process P3 request 1 unit of R3, is this request be granted? If yes, give a <sequence> in which all processes can run to completion
- (b) Write a short note to illustrate resource allocation graph with deadlock condition and without deadlock condition.

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OR

Q3

- (a) Define the difference between preemptive and non-preemptive scheduling. What advantage is there in having different time-quantum size on different level of a multilevel queuing system?
- (b) Why is it important to scale up system bus and device speeds as the CPU speed increases?
- (c) Differentiate Deadlock prevention and Deadlock avoidance.

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SECTION - II

Q4

- (a) Explain monolithic kernel and microkernel with proper diagram.
- (b) Distinguish internal fragmentation and external fragmentation. Provide any two solutions to avoid external fragmentation.
- (c) Explain the term 'Cycle Stealing' in the context of DMA operations.

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Q5

- (a) A computer has four page frames. The time of loading, time of last access and the R and M bits for each page is as shown below:

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Page Frame	Loaded	Last acc	R	M
0	126	280	1	0
1	230	265	0	1
2	140	270	0	0
3	110	285	1	1

- (a) Which page will NRU replace?
- (b) Which page will FIFO replace?
- (c) Which page will LRU replace?

- (b) Is the Process before and after the swap are the same? Give reason

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OR

Q5

- (a) Consider the following set of process in order P1, P2, P3, P4 and P5 with the length of the CPU burst time given in milliseconds. Their priorities are 3,1,4,5 and 2 respectively, with 1 being the highest priority. Calculate average waiting time and turn round time using following scheduling algorithms.

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- (1) Non-preemptive priority
- (2) Preemptive priority
- (3) SJF

Process	Arrival Time	Burst Time
P1	0	10
P2	0	1
P3	0	2
P4	0	1
P5	3	5

- (b) Write down the components of PCB.

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Q6 Answer the following questions (Any Four)

- 1) How an access matrix is used for implementing protection policies?
- 2) What is thread? Differentiate user-level thread and kernel-level thread.
- 3) What is the use of shell in the operating system? List various types of shell available in UNIX.
- 4) What is the cause of thrashing? How can the OS reduce the effects of thrashing and improve overall performance?
- 5) Explain Dining Philosophers problem.
- 6) Discuss linked and indexed allocation methods for free space on hard disk

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